

Task - 6: Introduction to Cryptography

Internship Task – Elevate Labs

This task focused on understanding the fundamentals of cryptography and how encryption, hashing, and digital signatures are used to protect data in real-world systems.

1. Introduction to Cryptography

Cryptography is the practice of securing information by converting it into an unreadable format. It ensures confidentiality, integrity, and authentication of data.

2. Symmetric and Asymmetric Encryption

Symmetric Encryption: Uses a single shared key for encryption and decryption. It is fast and commonly used for encrypting files. Example: AES.

Asymmetric Encryption: Uses a public key and a private key. It is used for secure key exchange and authentication. Example: RSA.

3. File Encryption Using AES

Files can be encrypted using AES to protect sensitive information. Only users with the correct key can decrypt and access the data.

4. RSA Key Generation

RSA keys consist of: - Public key - Private key

These keys enable secure communication and digital signatures.

5. Hashing and Data Integrity

Hashing converts data into a fixed-length value. Any change in the data results in a different hash value, helping verify integrity.

6. Digital Signatures

Digital signatures ensure data authenticity, integrity, and non-repudiation. They are widely used in secure communications.

7. Comparing Encryption Algorithms

Symmetric encryption is faster but requires secure key sharing. Asymmetric encryption is slower but safer for key exchange.

8. Real-World Applications

Cryptography is used in HTTPS, VPNs, secure messaging, and file protection.

Learning Outcome

This task strengthened my understanding of cryptography fundamentals and their real-world usage. I learned how encryption, hashing, and digital signatures help secure modern systems.